Activity Ch06

Using Random Forest Classification of Employee Productivity

Scenario:

A company has been tracking various performance metrics of its employees over time. The goal is to predict whether an employee's **productivity** will be **high** or **low** based on several factors such as **hours worked**, **tasks completed**, and other relevant attributes. The company wants to implement a **Random Forest Classification** model to accurately classify employee productivity and improve overall workforce efficiency.

As part of the project, students will develop a Random Forest model to predict employee productivity based on the provided dataset. The model will help the company focus on factors that are most indicative of high or low productivity, allowing for better resource allocation and training programs.

Objective:

The objective of this case study is to predict employee productivity using the Random Forest Classification algorithm. Students will analyze the factors contributing to productivity and use machine learning to classify employees into **high productivity** or **low productivity** categories.

Steps to Follow:

1. **Data Preparation:**
   - Load the dataset and clean it as necessary.
   - Analyze the variables to understand their impact on the target variable (productivity).
   - Create a new binary target variable for **Productivity** (1 = High, 0 = Low).
2. **Feature Selection:**
   - Identify relevant features (e.g., hours worked, tasks completed, etc.) that influence productivity.
   - Split the data into training and test sets.
3. **Random Forest Classification:**
   - Build a Random Forest model to classify productivity.
   - Evaluate the model's accuracy using cross-validation or other metrics (e.g., confusion matrix, F1-score).
   - Interpret the results to understand which features are most important in determining productivity.
4. **Model Interpretation and Business Recommendations:**
   - Based on the results, interpret the key factors affecting productivity.

- Provide recommendations on how the company can improve employee productivity (e.g., through training, better resource management, etc.).

Variables in the Dataset (Assumed based on a typical productivity dataset):

- **Hours Worked**: Number of hours worked by the employee in a week.
- **Tasks Completed**: The total number of tasks completed during the week.
- **Break Time**: The amount of time spent on breaks.
- **Overtime**: Whether the employee worked overtime (yes/no).
- **Project Complexity**: A rating of the complexity of the projects assigned to the employee.
- **Productivity**: The target variable indicating whether the productivity is **high (1)** or **low (0)**.

Questions for Analysis:

1. **Data Cleaning and Preparation:**
    - What data cleaning steps did you perform (e.g., handling missing values, removing outliers)?
    - How did you create the target variable for productivity classification?
2. **Random Forest Model:**
    - What are the key parameters you set for the Random Forest model (e.g., number of trees, depth)?
    - What is the model's accuracy in predicting employee productivity?
3. **Feature Importance:**
    - Which features have the most influence on predicting employee productivity, and how are they ranked by importance in the Random Forest model?
4. **Evaluation:**
    - How did the Random Forest model perform compared to other classification models (e.g., Decision Trees, Logistic Regression)?
    - What metrics (e.g., accuracy, precision, recall, F1-score) did you use to evaluate the model?
5. **Business Recommendations:**
    - Based on the model's results, what factors should the company focus on to improve productivity? For example, should they provide more training, reduce break times, or focus on project management?